

Subject:
Biochemistry

System:
Amino Acids and Proteins

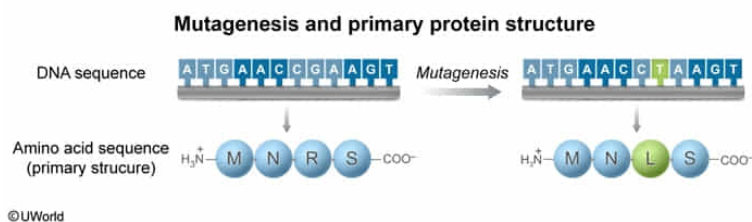
Which event could alter a protein's primary structure?

- ✓ ☒ A. Site-directed mutagenesis of the gene that encodes the protein [84%]
- ☐ B. A conformational change in the protein induced by ligand binding [3%]
- ☐ C. Protein denaturation by guanidinium chloride [8%]
- ☐ D. Reduction of disulfide bonds by β -mercaptoethanol [4%]

Correct

83% answered correctly

Explanation:



Proteins have multiple **levels of structure**: primary, secondary, tertiary, and quaternary.

- **Primary structure** is the **sequence of amino acids** from N-terminus to C-terminus, and is formed by **peptide bonds** between adjacent amino acids.
- **Secondary structure** consists mainly of alpha-helices and beta-sheets, and is formed by hydrogen bonds between the **backbones** of amino acids. It typically involves residues that are near each other in the primary structure.
- **Tertiary and quaternary structures** form through hydrogen bonds, salt bridges, hydrophobic interactions, and disulfide bonds between amino acid **side chains**. They yield the final, three-dimensional folded form of the protein. Tertiary structure

is formed by interactions within a single polypeptide chain, often between residues that are distant from each other in the primary structure. Quaternary structure involves interactions between separate polypeptide chains, called subunits, to form a multiprotein complex.

Because primary structure is the sequence of amino acids, it can only be disrupted by events that alter this sequence. These events include **cleavage of peptide bonds** by **proteases** and changes in the amino acid sequence by **mutations in the DNA** that encodes the protein of interest. Of the choices given, only **site-directed mutagenesis** affects the primary structure.

(Choice B) Protein interactions such as ligand binding or phosphorylation can lead to conformational changes that alter the three-dimensional folded state of a protein. However, this would not affect the sequence of amino acids that make up the protein.

(Choice C) Protein denaturation abolishes secondary, tertiary, and quaternary structure by unfolding the protein completely. However, denaturation does not disrupt peptide bonds, so it cannot change the sequence of amino acids (primary structure) of the polypeptide chain.

(Choice D) Disulfide bonds are involved in tertiary and quaternary structure, but not primary structure.

Educational objective:

Protein structure has four levels: primary, secondary, tertiary, and quaternary. Primary structure is the amino acid sequence from N-terminus to C-terminus, and can only be altered by breaking peptide bonds or by changing the identity of one or more amino acids within the original sequence (such as by mutation).

Time spent:0 sec

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